

Ivannov Kennedy Tatang

AI Application Developer | Full-Stack AI Developer

Selangor, Malaysia • [LinkedIn](#) • [daily.dev](#) • [Instagram](#) • [Personal Website](#)

Profile Summary

Full-Stack AI Developer specializing in modern web frameworks (Next.js/React) and Applied AI (Classic ML & NLP/LLMs). Experienced in engineering end-to-end intelligent applications with Python, FastAPI, and RAG pipelines—highlighted by an XMUM campus assistant with dynamic CMS knowledge management and an autonomous explainable AI trading agent using LangGraph and vector memory.

Technical Skills

- **Frontend:** React, Next.js, JavaScript, TypeScript, Tailwind CSS, Zustand, HTML5, CSS3
 - **Backend & Database:** Python, FastAPI, Node.js, REST APIs, Supabase (PostgreSQL, pgvector, RLS), MySQL, MongoDB
 - **AI & Machine Learning:** LLMs (Gemini, DeepSeek, GPT), LangGraph, LangChain, LlamaIndex, RAG, Prompt Engineering, scikit-learn, XGBoost, PyTorch, TensorFlow, NumPy, Pandas
 - **DevOps & Infrastructure:** Git, GitHub, Docker, Vercel, Render, Alpaca Paper Trading API & MCP, Linux
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Featured Projects

Full-Stack & AI Engineer — Alpaca XAI Agent: Explainable AI Trader

Aug 2026 • [lablab.ai Global Hackathon Submission](#) • [Project Link](#)

- Architected an explainable autonomous trading system using **LangGraph state machines** for schema-validated pre-trade thesis verification and automated execution via

Alpaca MCP.

- Built a post-trade autopsy pipeline using **Supabase pgvector** to embed trade outcomes and retrieve historical lessons during future evaluations, paired with an async 15-min background position auditor.
- **Tech Stack:** Next.js, Tailwind CSS, Python, LangGraph, Gemini API, DeepSeek API, Supabase (pgvector, RLS), Alpaca MCP

Full-Stack & AI Engineer — ShipSight: Early Warning & Decision Intelligence

Aug 2026 • **Devpost AI Builders Hackathon Submission** • [Project Link](#)

- Engineered a predictive logistics platform featuring a dual **XGBoost pipeline** (3.57d MAE Regressor, 91.6% accuracy Classifier) trained on 10,000+ USAID records, integrated with **TreeSHAP** for real-time root-cause risk driver visualization.
- Prompt-engineered **Google Gemini** into an agentic reasoning engine generating domain-grounded tactical mitigation playbooks, supported by an async **FastAPI** bulk manifest scanner and **Supabase PostgreSQL** active learning loop.
- **Tech Stack:** Next.js, Tailwind CSS, Python, FastAPI, XGBoost, TreeSHAP, Google Gemini API, Supabase (PostgreSQL)

Full-Stack & AI Engineer — XMUM CIKGU (Campus Assistant Platform)

May 2026 – Aug 2026 • [Project Link](#)

- Engineered a vector-backed **Retrieval-Augmented Generation (RAG)** engine (768-dim embeddings, **HNSW indexing**) to eliminate hallucinations in student Q&A retrieval.
- Developed a custom **Admin CMS with RBAC** allowing non-technical admins to update knowledge bases dynamically with automated real-time vector re-indexing.
- **Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, Python, FastAPI, Supabase (pgvector), Google Gemini API

Tech Lead & Web Dev Specialist — MARIO (Marine Route Optimizer)

May 2026 – Jul 2026 • [Project Link](#)

- Solved cloud deployment map rendering bottlenecks by decoupling heavy GIS map logic from backend responses (FastAPI coordinate payloads + Next.js Leaflet rendering).

- Engineered a real-time AIS simulation mode driven by trigonometric drift calculations to validate route optimization logic efficiently.
 - **Tech Stack:** Next.js, TypeScript, Tailwind CSS, Zustand, Leaflet.js, Python, FastAPI, Vercel, Render
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Academic Research & Recognition

Top 3 Peer-Voted Concept — Project \$VERA: Blockchain-Based Tax System

Xiamen University Malaysia • 2024

- Authored a conceptual research paper proposing a 3-layer RegTech framework integrating **Hedera Hashgraph** (Consensus), **ZK-SNARKs/STARKs** (Privacy), and **IoT Proof of Coverage** (Physical Truth).
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Education

Bachelor of Science in Artificial Intelligence Technology

Xiamen University Malaysia • 2025 – 2029

- **GPA:** 3.79 / 4.00 • **Concentration:** Machine Learning & NLP & FinTech
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Certifications & Languages

- **Alpaca AI Trading Agents Hackathon Certificate of Completion** — lablab.ai & NativelyAI (*Sep 2026*)
- **IELTS Academic Test** — Band 7.0 (*Nov 2024*)
- **Full-Stack & AI Workshops (Next.js & AI Recommendation APIs)** — Dunia Coding (*Feb 2025*)
- **Languages:** Indonesian (Native), English (Professional Working Proficiency)